ICA and Stochastic Optimization

Comparative Analysis and Creative Extension with EM-ICA

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Problem and Objectives

Blind Source Separation: recover independent sources $\mathbf{z}$ from observed mixtures

$$\mathbf{x} = W\mathbf{z}, \qquad V = W^{-1}, \qquad \mathbf{y} = V\mathbf{x}$$

Project Objectives:

- Implement 3 ICA algorithms: **Infomax**, **FastICA**, **SGD-ICA**
- Compare across multiple datasets with rigorous evaluation
- Propose a creative contribution: **EM-ICA** with learned score function

Key challenge: the score function $g(y) = \frac{d}{dy} \log p(y)$ must match the source distributions — but we don't know them in advance.

Unified Framework: One Gradient, Different Scores

Log-likelihood (course 5.6.2):

$$\ell(V) = \log|\det V| + \frac{1}{N}\sum_{i,j} \log p_j(v_j^\top x_i)$$

Gradient on whitened data ($V^{-\top} = V$, course 5.6.3):

$$\boxed{V \leftarrow V + \eta\left(V + \frac{1}{N}\, g(Y)X^\top\right)}$$

followed by symmetric decorrelation $V \leftarrow (VV^\top)^{-1/2}V$.

Score function choice:

- Super-Gaussian: $g(y) = -\tanh(y)$ (from $p \propto 1/\cosh(y)$)
- Sub-Gaussian: $g(y) = -y + \tanh(y)$ (Extended Infomax)

All gradient-based algorithms share this update — they differ only in how g is computed.

The Four Algorithms

Algorithm	**Method**	**Score**
Infomax Batch	Full-batch gradient	Kurtosis switching
FastICA	Fixed-point (Newton)	log cosh contrast
SGD-ICA	Mini-batch SGD	Kurtosis switching
EM-ICA	Stochastic EM	**Learned via MoG**

SGD-ICA uses Robbins-Monro decay $\eta_t = \eta_0/(1 + t/\tau)$:

$$\sum_t \eta_t = \infty, \quad \sum_t \eta_t^2 < \infty$$

FastICA (baseline) uses cubic-convergent fixed-point iteration — no learning rate, no online variant.

Question: can we do better than fixed kurtosis switching?

EM-ICA: Learning the Score Function

Motivation: kurtosis switching gives only 2 score shapes — it cannot distinguish sources with similar kurtosis.

Idea: model each source density as a Mixture of Gaussians:

$$p_j(z_j) = \sum_{k=1}^{K} \pi_{jk} \, \mathcal{N}(z_j \mid \mu_{jk}, \sigma_{jk}^2)$$

Algorithm (per mini-batch):

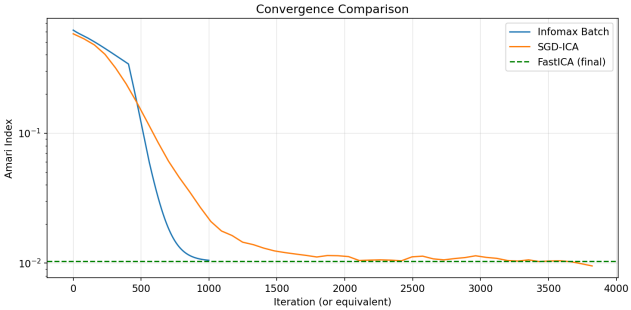
- **E-step:** compute responsibilities $r_{jk} = \pi_{jk}\mathcal{N} / \sum_{k'} \pi_{jk'}\mathcal{N}$
- **M-step (densities):** update (π, μ, σ^2) via EMA
- **M-step (unmixing):** use the learned MoG score

$$g_j(y) = \frac{d}{dy} \log p_j(y) = -\sum_k r_{jk} \frac{y - \mu_{jk}}{\sigma_{jk}^2}$$

  in the same gradient formula

The score now *adapts continuously* to the data.

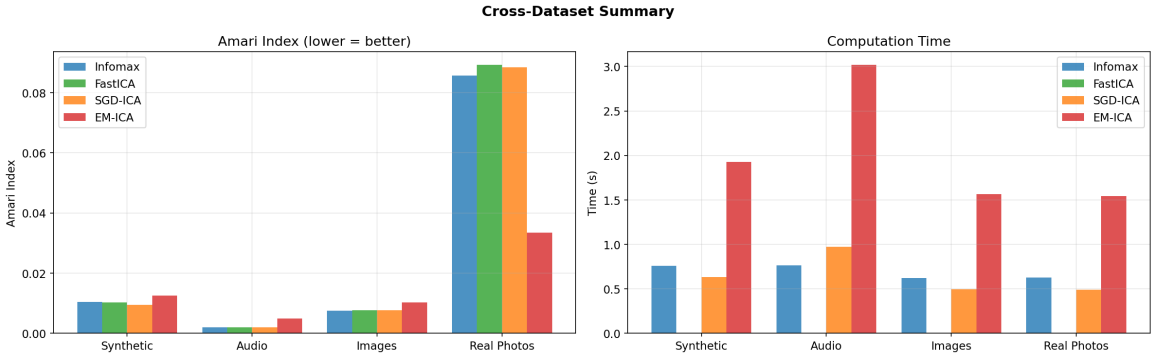
Convergence Behavior



- **FastICA:** cubic convergence (Newton)
- **Infomax batch:** smooth, $\sim$200 iterations
- **SGD-ICA:** noisier but consistent across runs

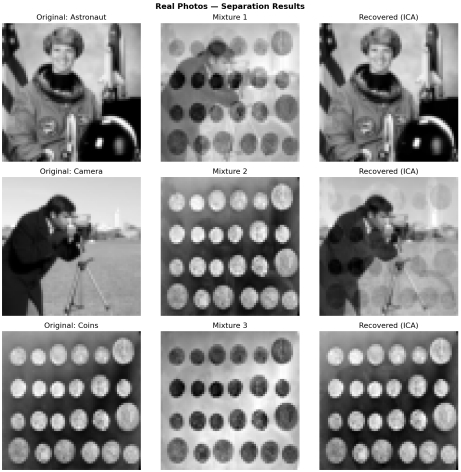
Stochastic noise in SGD acts as implicit regularization, helping escape local optima.

Cross-Dataset Visual Comparison



Amari index (left) and computation time (right) across all datasets

Real Photos: Visual Separation Result



Original sources $\rightarrow$ Mixtures $\rightarrow$ ICA recovery (EM-ICA, Amari $= 0.034$)

Results: Amari Index Across 4 Datasets

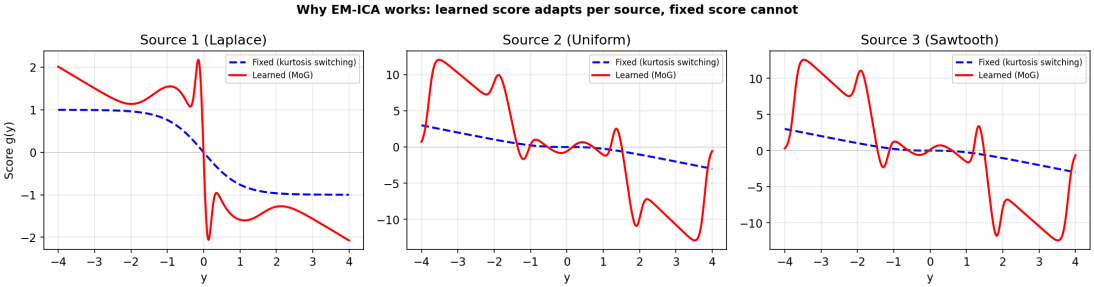
Dataset	Infomax	FastICA	SGD-ICA	EM-ICA
Synthetic	0.011	0.010	0.010	0.013
Audio-like	0.002	0.002	0.002	0.005
Synth. Images	0.008	0.008	0.008	0.010
Real Photos	0.086	0.086	0.087	**0.034**

Key observations:

- All algorithms work well on synthetic / audio / image data (Amari < 0.02)
- **EM-ICA wins on real photos: 2.5$\times$ better**
- FastICA is $\sim$500$\times$ faster (Newton vs. gradient)

Why? On real photos, all 3 sources have $\kappa \approx -1.2$. Kurtosis switching cannot distinguish them, but the MoG learns the actual density shapes.

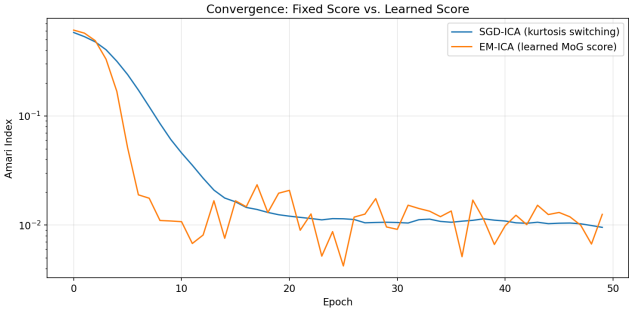
Why EM-ICA Works: Score Function Comparison



- **Blue dashed:** fixed score (kurtosis switching) — identical for sources 2 and 3
- **Red:** learned MoG score — different shape per source

Insight: when sources differ in density shape but not in kurtosis, learning the score is necessary.

EM-ICA Convergence: Learned vs. Fixed Score



- **SGD-ICA:** smooth descent with fixed score
- **EM-ICA:** noisier — the MoG parameters and V are learned **simultaneously**
- Both reach comparable Amari at convergence

The extra noise is the price of flexibility: EM-ICA learns the densities on the fly.

Discussion and Limitations

Key findings:

- **Score function is critical** — using $-\tanh$ alone fails on sub-Gaussian sources
- **Gaussian sources fail completely** (Amari > 0.3) — confirms identifiability theorem
- **Infomax batch has high variance** (0.036 ± 0.076) — some runs don't converge
- **EEG artifact detection:** all algorithms identify a high-kurtosis component ($\kappa \approx 45$, likely eye blinks)

Limitations:

- EM-ICA has $3KD$ extra parameters — slower per iteration
- MoG score is noisy in low-density tail regions
- Only 3 EEG channels tested (clinical EEG uses 64+)

Conclusion

Summary:

- Implemented 4 ICA algorithms in a **unified gradient framework**
- Tested on **5 datasets** with rigorous evaluation (10 runs, error bars)
- Confirmed ICA identifiability theorem experimentally

Main contribution: EM-ICA with learned MoG score

- Same gradient as Infomax/SGD — only g changes
- **2.5**$\times$ **better** on real photographs (Amari 0.034 vs. 0.086)
- Particularly effective when sources have similar kurtosis

Future directions: natural gradient, normalizing flows, VAE connection.

Thank you!